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Waterless leather dyeing with dense carbon dioxide as solvent for dyes

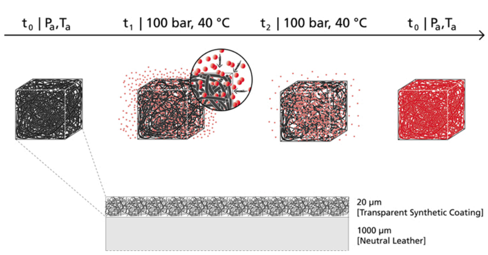
M. Prokein [a,*], T. Dyes [a], M. Renner [a], E. Weidner [a,b]

[a] Fraunhofer Institute UMSICHT, Division Products, Osterfelder Str. 3, 46047 Oberhausen, Germany
[b] Ruhr University, Chair of Process Technology, Universitätsstr. 150, 44780 Bochum, Germany

HIGHLIGHTS

- Waterless leather dyeing by $scCO_2$ at 100 bar and 40 °C leads to intense and uniform colorizations of the leather surface.
- The dye exhaustion is almost 100%.
- $ScCO_2$ dyeing saves 90% of dyes.

GRAPHICAL ABSTRACT



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ABSTRACT

Conventional leather dyeing consumes high amounts of dyes and increases pollution of tannery wastewater. We have focused on the reduction of dyestuff and the avoidance of water and additional chemicals for the colorization of leather. Waterless leather dyeing with supercritical CO_2 as solvent for the dye 1-(methylamino) anthraquinone was performed in a technical scale autoclave. The intensity and uniformity of the dyed leathers was assessed using spectroscopic color measurements. The rub fastness was assessed according to DIN EN ISO 11640. In the result, waterless dyeing at 100 bar and 40 °C leads to uniform, intensive colorizations of the leather surface at a usage of 0.1 g of dye per square foot of leather. The dyed leathers achieve the highest rub fastness grades. The exhaustion of dyestuff is almost 100%. The waterless dyeing process offers potential to save significant amounts of dyestuff and reduce wastewater emissions by 100%.

1. Introduction

1.1. The economic and ecological side of leather

The raw materials for the tanning industry are hides and skins, of which over 99% are derived from animals raised primarily for wool, milk and/or meat production. This fact clearly illustrates the ecological role of tanneries; recovering a by-product, which in the absence of the leather industry would have to be disposed of [1].

Leather is commercialized worldwide to make products for various applications. More than 50% of all leather is used in shoe production. Upholstery and clothing constitute further large markets with shares of about 20% and 17% respectively [2]. China and Europe are the two largest leather-producing regions. The annual turnover of the leather industry including leather products is more than 700 billion euros [3,4].

In the leather production process, converting 1 ton of raw hide into 200 kg of leather through various process steps uses up to 50 m^3 of water and about 500 kg of chemicals. About 20% of the chemicals remain in the finished leather product while 80% remain in the wastewater. Therefore, tannery wastewater incurs enormous wastewater treatment

* Corresponding author.
E-mail address: michael.prokein@umsicht.fraunhofer.de (M. Prokein).

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costs as well as causing environmental pollution in some areas of the world [5].

1.2. Conventional leather dyeing

Dyeing of leather is a central manufacturing step in leather production. The colorization determines later use, as the surface of the leather is subject to correspondingly quality requirements [6,7].

The conventional dyeing process usually consists of two steps: dyeing the cross-section of the collagen (natural collagen from animal hides is the basic material for leather products) in an aqueous solution and coating the leather surface with water-based binders containing dyes or pigments. The binders that can contain colorizing agents are divided into the basecoat and topcoat. Both coatings serve various functions. The main function of the basecoat is to compensate for irregularities on the surface of the leather. The topcoat refines the surface. An additional transparent coating can optionally be applied to avoid rub off for different leather products. Applying the coatings is the final step in leather production to prepare the surface for the product's area of application e. g. shoes, bags, clothes, seats etc. These steps are collectively referred to as finishing [8–10].

The amount of water, dyes and chemicals used to achieve specific colors varies considerably. About 5 m^3 of water, 150–350 kg of dyes and over 100 kg of chemical agents such as surfactants, acids, bases, levelling agents, chelating agents, electrolytes, emulsifying and softening agents are applied to 1 ton of animal hide to meet the high colorization requirements [8,11,12,13]. According to estimates by modern tanneries, more than 30% of the dyes used to colorize the cross-section are lost with the wastewater. In finishing, depending on the type of leather and thickness of the coating, over-spraying causes dye or pigment losses of between 15% and 40% [14]. A further reason for dye losses is that the finished leather is cut to size after conventional dyeing at the end of the leather production process. This means that about 15% of leather that has already been dyed has to be disposed of as solid waste [5]. The production of leather dyes contributes to the ecological critical impact categories for leather products. The use of dyes to colorize leather needs to be reduced to make the leather industry more sustainable [15].

Furthermore, leather is a difficult substrate to dye to a level and consistent shade. Therefore, the complex conventional dyeing process cannot react flexibly to customer requests with regard to the coloring [16].

Waterless dyeing using dense CO_2 as a solvent for dyes or pigments is a promising approach to save dyes and chemicals, avoid wastewater contamination and make tanneries more flexible. Since the cost of dyes at tanneries usually accounts for more than 20% of their total costs for chemicals, the loss of colors has a significant influence on production costs.

1.3. Supercritical CO_2

1.3.1. The properties of dense CO_2

CO_2 is naturally occurring, chemically inert, physiologically compatible, inexpensive, and readily available for industrial consumption. Depending on the pressure and temperature, CO_2 appears in solid, liquid, gaseous and supercritical states. The states of CO_2 set its diversity of properties. Table 1 shows the order of magnitudes of the density, viscosity and diffusion coefficients of CO_2 depending on the physical state [17].

The ability of CO_2 to act as a solvent is poor or non-existent in its gaseous state. The reason for this is its low density in the order of magnitude of 10^{-3} g/cm^3. In its liquid and supercritical states, the solvent properties improve significantly. This makes liquid and especially $scCO_2$ interesting for several extraction and impregnation processes [18]. Colorization of a material by dissolving a dye or pigment in CO_2 and transporting it into the matrix of the material as an impregnation process is the focus of this work.

Liquid CO_2 can be converted to a supercritical fluid by increasing the temperature and pressure simultaneously. At the same pressure, the density of liquid CO_2 and thus its dissolving power is higher compared with $scCO_2$. However, the density of $scCO_2$ and its dissolving power can reach the order of magnitude of liquids by increasing the pressure. In comparison with the liquid state, the use of $scCO_2$ usually offers a number of advantages. The viscosity of a supercritical fluid is in the range of gases. This has a decisive influence on the convective mass transport. Low viscosity promotes intensive mass transport and ensures a constant dye concentration in the fluid phase, preventing it from being consumed locally. Furthermore, the diffusion coefficients of supercritical fluids are also in the range of gases, which promotes the introduction of dyes into porous and fine structures such as textiles or leather. The surface tension of CO_2 is negligible in the supercritical state. This enables optimal wetting of surfaces. The simple separation of CO_2 and the dissolved substance by lowering the pressure is also considered to be a major advantage [17,19].

The critical pressure of a substance is seldom below the ambient pressure. Accordingly, procedural use of supercritical fluids essentially always requires a pressure vessel and the associated peripherals. When using CO_2, the technical effort remains moderate. The critical pressure and temperature required to reach the supercritical state of CO_2 are comparatively low. The critical pressure of CO_2 is 73.8 bar and the corresponding critical temperature is 31.1 °C [18].

For the impregnation of synthetic polymers, CO_2 allows changes in the degree of plasticization and swelling of amorphous and semi-crystalline materials. The swelling improves the ability of polymers to absorb dyes that are dissolved in CO_2. The plasticization of the amorphous phase in semi-crystalline polymers promotes the mobility of the polymer chains, which can be rearranged into more ordered configurations. This results in induced crystallization and concomitant changes in the morphology. The higher the pressure under isothermal conditions, the higher the CO_2 uptake of the material [20,21].

1.3.2. Dyeing with supercritical CO_2

$ScCO_2$ dyeing of polyester textiles has been known for more than 20 years and the process has been investigated in detail by many researchers [21–23]. The DyeCoo company has implemented the technology industrially and dyes polyester fibers in Taiwan and Thailand. The process operates at pressures of about 250 bar and at 120 °C. Despite the apparently high process conditions, $scCO_2$ polyester dyeing reduces chemical usage and energy consumption by 50% respectively compared to water-based coloring. Furthermore, the waterless process saves 25 liters of water per dyed T-shirt [24].

In contrast to synthetic polyester fibers, $scCO_2$ dyeing of natural materials as collagen is still challenging and has not been commercialized [25]. The polar character of the natural fibers requires very different dyestuffs and conditions in comparison with synthetic fibers [26]. Dye classes that are usually used for natural fibers, such as reactive, direct or acid dyes, are almost insoluble in CO_2 [27]. It is also assumed that CO_2 is unable to sufficiently open hydrogen bonds in order to facilitate the diffusion of the dye in natural materials [21]. Despite the challenges, many researchers are investigating different approaches to $scCO_2$ dyeing of natural materials. The goal is to make textile dyeing more economical and to reduce the environmental burden of water-based dyeing technologies. One approach is to add an impregnating agent to natural fibers. In the agent, the color can be redeemed in

Table 1
Comparison of important properties of gaseous, liquid and supercritical carbon dioxide [17].

	Gas	Liquid	Supercritical
Density [g/cm^3]	10^{-3}	1	0.6
Diffusion coefficient [cm^2/s]	10^{-1}	5 10^{-6}	10^{-3}
Viscosity [g/cm*s]	10^{-4}	10^{-2}	10^{-4}

the subsequent CO_2 dyeing process [28]. In order to use water-soluble dyes with a high affinity against natural fibers, the use of cosolvents was considered or corresponding micro emulsions were generated [29, 30]. In addition, the chemical modification of fibers using hydrophobic functional groups to enable interactions with disperse dyes was focused on by different scientists [31]. However, the approaches described only achieved colorizations of natural fibers using $scCO_2$ as a solvent on a laboratory scale. This means that satisfactory colorations were always associated with chemical-intensive pre-treatment. The advantages of the technology did not outweigh the disadvantages of the pre-treatment.

One approach to adapting CO_2 dyeing without pre-treatment of the fibers is the synthesis of CO_2-soluble, reactive disperse dyes to bind the dye to the fibers chemically. This would keep the technical effort of the dyeing process low [21,25,32,33]. However, CO_2-soluble reactive dyes are not commercially available yet [25]. Consequently, CO_2-based coloring of natural substrates has not been implemented industrially.

Campardelli et al. report on a process called "SuperLip" for leather dyeing. In the process, $scCO_2$ is used to encapsulate dyes in liposomes for the colorization of leather. However, in addition to $scCO_2$ and dyes, different components such as water and lipids have to be used [6].

Waterless CO_2 dyeing of natural collagen using commercialized CO_2-soluble dyes results in unsatisfactory colorizations [34]. However, during the leather production process, different leather chemicals modify and functionalize the natural collagen [35]. In particular, the leather surface is usually refined by coating it with different mixtures of finishing agents [9]. The synthetic character of these agents include the potential to show a higher affinity to CO_2-soluble dyes than natural collagen. However, in the leather production process, the maximum operating temperature is usually below 60 °C. Chromium-tanned leather reaches the highest thermal stability and is resistant to a temperature of about 100 °C. At a commercial polyester dyeing temperature of 120 °C, the leather area would decrease and reduce the value of the product. [11,36].

Literature on waterless CO_2 dyeing of leather using pure dyes is not available at present.

1.4. Research issue

The aim of this work is to examine waterless leather dyeing by colorizing the leather coating with a CO_2-soluble dyestuff in a dense CO_2 atmosphere. We investigated whether it is possible to dye the entire surface of leather coated with a mixture of different finishing agents, depending on different pressures of between 80 and 250 bar and temperatures of between 20 and 80 °C. Furthermore, we considered the influence of the amount of dyestuff in relation to the leather area and loading of the dyeing autoclave.

In comparison with conventional leather finishing, excess dyes and pigments that contaminate the wastewater due to over-spraying should be minimized.

2. Materials and methods

2.1. Materials

The commercially available Solvaperm Red PFS CO_2-soluble dye from Clariant Deutschland GmbH was used. The chemical name of the dye is 1-(methylamino)anthraquinone. CO_2 with a purity greater than 99.9% was supplied by Nippon Gases Deutschland GmbH. Heller Leder GmbH & Co KG provided neutral furniture leathers. Neutral leather is defined as leather processed without dyestuffs or finishing. The LCH color space values (Section 2.2.3) of the leather are: $C^* = 2.2$, $L^* = 76.5$, $H^* = 191.5$. The thickness of the leather was 1 mm. Langro GmbH & Co. KG provided the recipe and finishing agents for the basecoat and topcoat. The recipe and the components for the coatings are listed in Table 2.

2.2. Methods

2.2.1. Leather preparation

A Lederfabrik Heinen GmbH & Co. KG industrial spraying plant coated neutral leather sides with the full recipe consisting of the basecoat and topcoat (Table 2). Five leather sides were passed through the coating system and a dryer several times. 2 g of basecoat or topcoat per square foot of leather were applied with each pass. The basecoat was applied in four runs and the topcoat in two runs.

The neutral leathers were divided into samples of 0.20 ft. by 0.33 ft. for the dyeing experiments in autoclave 2 (Section 2.2.2). The size allowed loading of the rotating basket with sufficient freedom of movement and mixing of the samples during the dyeing process. The set dimensions of the samples were sufficient for all analytics considered. Circular samples with a diameter of 0.07 ft. were prepared for the dyeing experiments in autoclave 1 (Section 2.2.2).

2.2.2. Supercritical CO_2 dyeing

The dyeing test series were carried out in a 63 mL (autoclave 1) and a PLC-controlled 5 L autoclave (autoclave 2).

The high-pressure view cell enabled investigation at operating temperatures of between 20 °C and 80 °C and pressures of up to 250 bar. The maximum temperature of the 5 L machine was limited to 50 °C. However, the machine allowed the processing of 5.8 ft^2 of leather per batch in a rotating basket.

Fig. 1 shows a schematic diagram and photograph of autoclave 1. A high-pressure pump compresses CO_2 that is stored at 60 bar in a CO_2 tank to pressurize the cell. By opening valve 3, the cell can be manually pressurized to the desired pressure. Valve 4 enables the depressurization. A sapphire window at the front of the view cell allows the process to be observed. A lamp installed behind a second rear window lights up the interior of the cell. A stirrer can optionally improve the mass transport.

For each trial, three neutral leather samples were placed in the high pressure view cell. In addition to this, 0.01 g of solid dyestuff were added. The dyestuff amount was then in large excess compared with the solubility of Solvaperm Red PFS in dense CO_2 [37] and with the mass of the applied coating. Afterwards, the cell was closed, preheated to the desired temperature for 30 min and pressurized to the desired pressure. After 10 min the pressure and temperature were constant. The residence time at constant parameters was set to 1 h. At the end of the process, the pressure was released at 5 bar per minute and the colorization was assessed as described in Section 2.2.3.

Fig. 2 shows a schematic diagram and a photograph of autoclave 2. A horizontal, cylindrical autoclave includes a rotatable basket. The basket allows the mixing of leather samples during the dyeing process. Motor M speeds up the basket to 56 rounds per minute. CO_2 is stored in a reservoir at 60 bar. The autoclave can be pressurized up to 250 bar. The machine is able to operate at temperatures between 35 and 50 °C. Between outlet valves V3 and V4, a cyclone for the separation of solids and liquids is installed.

Table 2
Composition of the Basecoat and Topcoat.

Basecoat		Topcoat	
Component	Quantity [g]	Component	Quantity [g]
Water	400	Water	400
Telaflex U 44 (U44)	150	Telaflex U 44 (U44)	150
Telaflex U 410 (U410)	100	Telaflex U 410 (U410)	150
Langrofin VLM 29 (VLM29)	50	Langrofin VLM 29 (VLM29)	50
Telaflex A 118 (A118)	100	Telaflex U 500 (U500)	100
Telaflex U 529 (U529)	50	Langrofin GW 5 (GW5)	50
Langrofill 5 (Fill5)	50	Langrofin GW 78 (GW78)	50

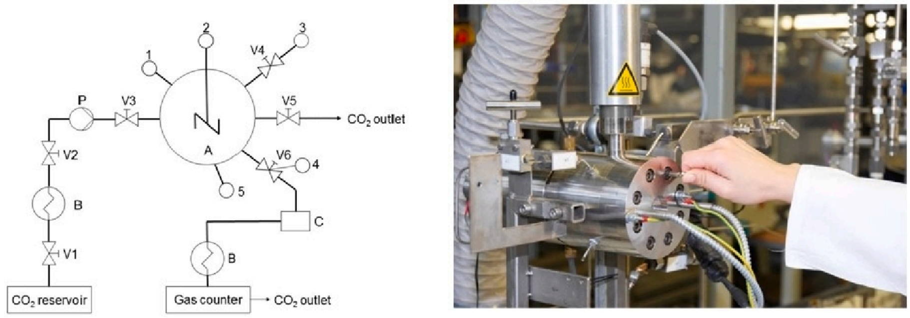

Fig. 1. Schematic diagram and photograph of the 63 mL high-pressure view cell; A: Cell, B: Heat exchanger, C: Separator, P: Pump, V: Valve, 1: Pressure sensor, 2: Stirrer, 3: Inlet for liquids, 4: Temperature sensor, 5: Temperature control.

In contrast to the trials in autoclave 1, dyestuff was not only used in excess. For the test series discussed in Section 3.2.2, defined dyestuff amounts related to the leather area were used. The amounts of dye without excess are described in the corresponding sections. The autoclave was preheated to the desired temperature. After the dye and leather samples were place in the horizontal basket, the PLC-controlled dyeing process was initiated. The pressurizations to the desired pressures took about 10 min. The residence time at constant parameters was 1 h. At the end of the process, the pressure was released at 5 bar per minute. The basket rotated during the procedure at a speed of 30 rpm.

2.2.3. *Color measurements*

Color measurements were conducted using a CAS 140CT-154 spectrophotometer from Instrument Systems Optische Messtechnik GmbH. The spectrophotometer analyzes a spectral range from 220 nm to 1020 nm at a resolution of 3.7 nm. A halogen lamp (λ = 380–2200 nm) serves as a light source and is clamped in the RMH 45 reflection probe.

Fig. 3 shows the setup of the color measurement system schematically. For the measurements, a reference or a leather sample (7) is placed below the housing (6). The halogen light (1) irradiates the sample at an angle of 45° (3). The optical fiber receives light orthogonally radiated from the sample (4). This ensures that over-illumination by directly reflected light is avoided and only diffusely reflected light enters the measuring device (5).

The photometric data obtained was evaluated using the device-independent CIELAB color space as standardized by Euronorm EN ISO 11664-4. The parameters L* (lightness), a* (color gradient varying from green to red) and b* (color gradient varying from blue to yellow) describe a specific color in the three-dimensional color space. The measurement signals and conversion into the L*a*b* color space were evaluated using the SpecWinPro software supplied by the manufacturer of the spectrophotometer.

In this work, the L*a*b* color space was converted in cylinder coordinates according to formulas (1) and (2). In the resulting L*C*H* color space, the lightness L* is retained as the longitudinal axis of the cylinder. C* indicates the saturation (chroma) as a radius to the origin and H* describes the hue angle. Fig. 3 shows the correlation between parameters a*, b*, H* and C*.

$$H^{\circ}_{ab} = \cot\left(\frac{a^*}{b^*}\right) \tag{1}$$

$$C^*_{ab} = \sqrt{a^{*2} + b^{*2}} \tag{2}$$

The spectrum under consideration is limited to a wavelength range of 380–780 nm. This corresponds to the area perceptible to the human eye. Average values with standard deviations of at least three samples per trial with three measurement points of each colored sample are calculated. The photometric data of the leather samples are related to an industrial white reference.

Since the CO_2-soluble dye used cannot be processed in aqueous

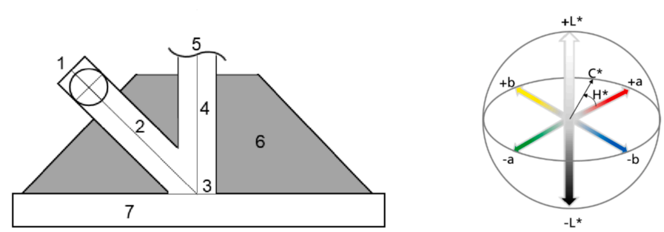

Fig. 3. Schematic diagram of the setup of the spectrophotometer (left) to determine the Lab color space (right [38]), 1 Halogen lamp, 2 Light, 3 Reflection, 4 Orthogonal reflected light, 5 Detector, 6 Housing, 7 Leather sample.

Fig. 2. Schematic diagram and photograph of the 5 L high-pressure dyeing autoclave, B: Heat exchanger, P: Pump, M: Motor, Z: Cyclone, V: Valve.

solutions, there are no comparative LCH color space values for leathers that are dyed with Solvaperm Red PFS. Furthermore, conventional red leathers are produced in a variety of hues. Therefore, there is no reference standard for red leathers. To assess the colorization of the CO_2 dyeing process, we compare the resulting chroma and hue angles with color values of different colorized polymers. Table 4 shows C* and H* values of four materials dyed with Solvaperm Red PFS [39]. Lightness values for the dye used are not available. In addition, we compared the resulting color values with the initial color values of the neutral leather.

2.2.4. Rub fastness test

The test method according to DIN EN ISO 11640 was performed using a Bally Finish Tester 9001. After dyeing, each sample was clamped in the test device and stretched by 10%. A prepared felt specimen was placed in the stamp of the test device, weighted with 100 g and then moved back and forth on the sample for 40 cycles. Table 3 shows the conditioning of the felts. The rub fastness can be assessed by a subsequent comparison of the samples using a grey-scale according to ISO 105-A02. The measurement scale contains values between 1 and 5. A value of 1 corresponds to complete coloration of the pelt. A value of 5 represents the highest quality, because no rub off is observed.

3. Results and discussion

3.1. CO_2 dyeing in autoclave 1

The influence of the pressure between 80 bar and 250 bar and temperatures between 20 °C and 80 °C was investigated in autoclave 1. In all experiments that resulted in colorization of the leather, the dyestuff was selectively absorbed by the coating. That means that the colorization of the natural leather as a substrate of the coating was low. As an example, Fig. 4 shows a photograph of the coated surfaces and uncoated sides of leather as samples before and after CO_2 dyeing at 100 bar and 40 °C. The color of the coated sides before dyeing (1) and after dyeing (3) deviate significantly from each other because of the absorbed dyestuff. The change in the color of the uncoated sides before CO_2 dyeing (2) and after CO_2 dyeing (4) is due to precipitated dyestuff and is not caused by impregnation of the dyestuff into the leather matrix.

3.1.1. Influence of the pressure and temperature on the chroma and lightness

Figs. 5–7 show the LCH values of the leather coatings depending on the pressure and temperature. However, the parameter sets of 80 bar at 60 °C, 80 bar at 80 °C and 100 bar at 80 °C did not result in a change in the initial LCH values of the coatings (C* = 2.2, L* = 76.5, H* = 191.5 (Section 2.1)) and are not included in Figs. 5–7.

By comparing Figs. 5 and 6, we observe that the L* values correlate to the C* values. This means that a pressure-induced or temperature-induced increase in the chroma leads to a decrease in the lightness.

At 20 °C and 80 bar, the chroma value shown in Fig. 5 reached 25.6% with a standard deviation (SD) of 0.9%. By increasing the pressure from 80 bar to 250 bar at 20 °C, the C* value increases to 43.2% (SD = 2.4%).

By increasing the temperature from 20 °C to 40 °C, the C* value at 80 bar of 12.5% (SD = 5.4%) is lower than at the lower temperature and

Table 3
Condition of pelts for rub fastness tests.

Condition	Ingredients
Dry	–
Wet	1 g deionized water
Perspiration	1 g artificial perspiration 100 mL of perspiration consist of: • 99.4 g deionized water • 5 g sodium chloride • 5.3 g ammonium hydroxide

Table 4
Color values of different materials dyed with Solvaperm Red PFS [39].

Material	Polystyrene	ABS	PC	PETP
Chroma [%]	65.1	59.6	58.5	55.2
Hue [°]	24.5	13	15.9	9

Fig. 4. Photograph of coated surfaces and uncoated backs of the sides of leather used as samples with a diameter of 0.07 ft before and after $scCO_2$ dyeing at 100 bar and 40 °C in high-pressure autoclave 1, 1 coated side before dyeing, 2 Uncoated side before dyeing, 3 Coated side after dyeing, 4 Uncoated side after dyeing.

the same pressure. By increasing the pressure to 100 bar, the C* value increases to 53.3% (SD = 1.7%). A further increase in the pressure and temperature does not noticeably influence the chroma values.

Complementary to the chroma, Fig. 6 shows the lightness values of the coatings depending on the dyeing pressure and temperature. At 80 bar, the L* values are 66.1% (SD = 0.5%) at 20 °C and 71.9% (SD = 2.2%) at 40 °C. By increasing the pressure to 100 bar at 40 °C, the L* value of 53.2% is within the SD of the lowest lightness values. Thus, increasing the temperature to 80 °C and the pressure to 250 bar does not lead to significant changes in the L* values.

The trends in the chroma and lightness values as described show that it is possible to incorporate a dyestuff into the matrix of the polymer coating using dense CO_2. The initial C* value of the neutral leather increased from 2.2% to more than 50% and is in the range of polymers that were dyed with Solvaperm Red PFS (Table 4).

In contrast to the coating, the substrate of the coating showed no colorization. This is because the substrate of the polymer coating consists of polar collagen fibers and mainly polar leather chemicals as described in Section 1.2. The polar character of the material prevents absorption of the low polar CO_2-soluble dye [40,41].

The parameter sets of 80 bar at 60 °C, 80 bar at 80 °C and 100 bar at 80 °C did not result in a change in the initial LCH values. Furthermore, the parameter sets of 80 bar at 40 °C and 100 bar at 60 °C lead to only slight changes in the initial LCH values. This is explained by the pressure-dependent and temperature-dependent solubility of dyes in CO_2. The ability of a dyestuff to dissolve in CO_2 is a basic criteria for CO_2 dyeing [24]. The concentration of dyes dissolved in CO_2 increases along with the CO_2 density [21,42,43]. The density of CO_2 increases by increasing the pressure or decreasing the temperature. At 80 bar and

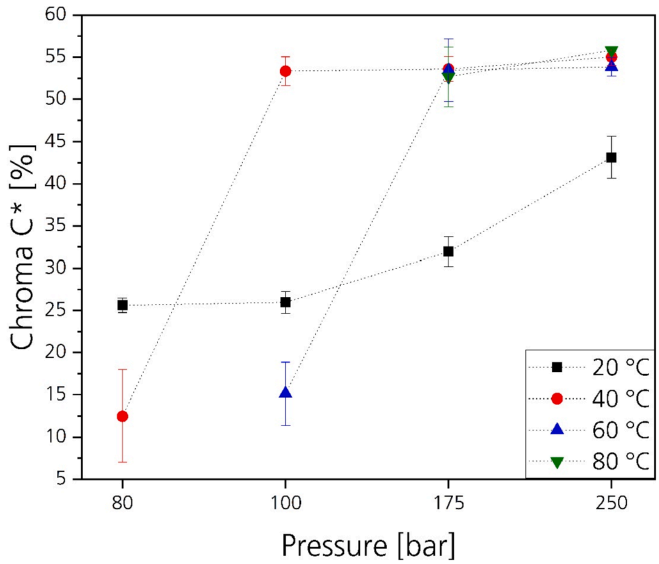

Fig. 5. Influence of the pressure and temperature on the chroma of leather coatings after CO_2 dyeing with Solvaperm Red PFS in autoclave 1.

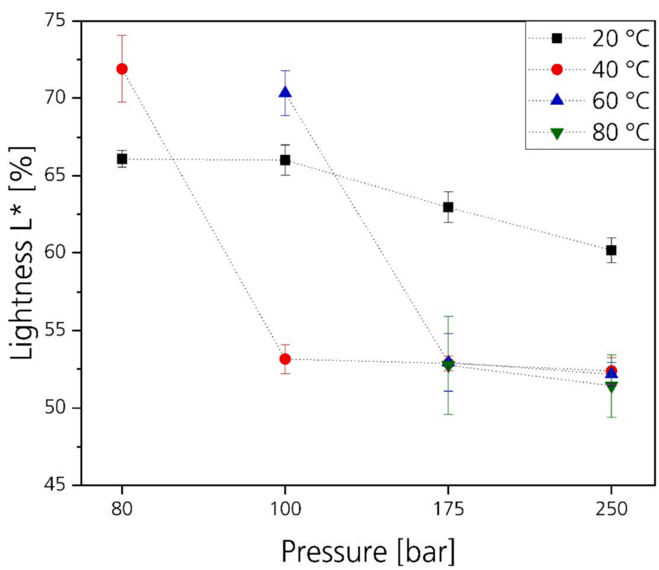

Fig. 6. Influence of the pressure and temperature on the lightness of leather coatings after CO_2 dyeing with Solvaperm Red PFS in autoclave 1.

20 °C, the density is 830 kg/m^3, which is three to five times higher compared with the density at 40 °C ($\rho = 278$ kg/m^3) and 80 °C ($\rho = 160$ kg/m^3) [44]. The resulting dye concentration at 20 °C causes the highest chroma while, at the higher temperatures, the lower or non-existent dye concentrations lead to poor or no colorization.

We assume that the mechanism of colorization of the polymeric leather coating with the low polar dye molecular is physical absorption. The increase in chroma values at 20 °C by increasing the pressure shows that the higher the density under isothermal conditions, the higher the CO_2 uptake of a polymer, as already observed by other authors [21]. However, at 100 bar, the highest chroma of more than 50% is reached at 40 °C and not at the highest CO_2 density, which occurs at 100 bar and 20 °C. This shows that the highest density and therefore dye concentration does not inevitably correlate to the highest dye uptake and colorization. Furthermore, the diffusivity significantly influences the rate of physical absorption [21,45,46]. At 100 bar and 20 °C CO_2 is liquid, while at 100 bar and 40 °C CO_2 is above its critical temperature

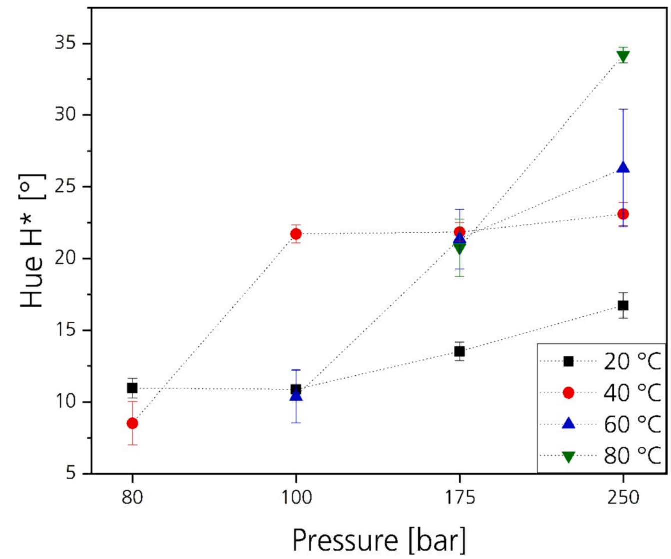

Fig. 7. Influence of the pressure and temperature on the hue angles after CO_2 dyeing of leather coatings with Solvaperm Red PFS in autoclave 1.

and critical pressure (Section 1.3.1). The improved diffusivity in supercritical condition compared with liquid condition is mainly responsible for the improved dye uptake [17]. It is worth noting that at 80 bar and 40 °C CO_2 is already above its critical pressure and temperature. However, as already discussed, it can be assumed that the density at these parameters of 278 kg/m^3 limits the dye concentration and colorization. This shows that the combination of pressure and temperature is essential to achieve a sufficient density to dissolve dyestuff and a high diffusivity to load the coating with dyestuff.

In CO_2 dyeing, dyes are able to diffuse in amorphous and semi-crystalline phases, whereby amorphous phases are advantageous for the mass transport [47]. This is due to the chain mobility of the matrix that usually increases by increasing the temperature [50]. Higher temperatures at a constant pressure facilitate the diffusivity of CO_2 in synthetic polymers [48,49]. In this context, it is surprising that at more elevated process parameters the coloration of the leather coating does not become more intense. The dye uptake already results in a range of typical C* values for Solvaperm Red PFS of more than 50% at 100 bar and 40 °C. Because an increase in temperature at supercritical conditions does not lead to an increase in the C* value, we assume that the coating consists mainly of amorphous phases. Those phases enable the dye to diffuse already at 40 °C and makes it suitable for CO_2 dyeing at moderate temperatures. The optimal CO_2 dyeing parameter for the coloration of leather coating is low in comparison with CO_2 dyeing of polyester at 250 bar and 120 °C [50].

We note that the coating consists of different finishing agents containing different polyurethanes, acrylates and further components. For a detailed explanation of the mass transport in the matrix of the coating, the structures of the single finishing agents have to be considered in additional work.

3.1.2. Influence of the pressure and temperature on the hue angles

Fig. 7 shows the hue angles depending on pressures between 80 bar and 250 bar and temperatures between 20 °C and 80 °C. The H* value trends that resulted at 20 °C and 40 °C concur with the previously discussed C* value and L* value trends. At 20 °C, the H* values increased by increasing the pressure. In addition, the H* values are below the H* values that resulted at 40 bar and the pressures above 80 bar considered. The H* values of about 22° that resulted at 40 °C and pressures between 100 bar and 250 bar are assigned to an intensive red color. The initial hue angle of 191° assigned to blue is completely covered by the red dye. As for the C* values and L* values, the H* values are constant at 40 °C and between 100 bar and 250 bar.

At 60 °C and 80 °C, the H* value trends deviate from the C* value and L* value trends. In the range of 175–250 bar, in which the chroma was constant, the H* values increased. Especially at 250 bar and 80 °C, the H* value of 35° is unusual for red dye (Section 2.2.3). The hue angle of about 35° that resulted at 250 bar and 80 °C is assigned to orange. The orange color occurs because of partitioning of dye between the CO_2 phase and the polymer matrix. The partitioning is cause by a portion of dyestuff not being absorbed by the coating but rather being precipitated and adsorbed on the leather surface. Precipitated dyestuff due to unfavorable partitioning of dye at various parameters has already been observed by other researchers [21]. Kazarian et al. suggest that a high affinity of a dye relative to the fluid is key to identifying optimal CO_2 dyeing conditions [42].

3.2. $ScCO_2$ dyeing in autoclave 2

3.2.1. Influence of pressure, temperature and rotation

The previous test series shows that colorization of coated leather is possible using subcritical and supercritical CO_2. However, the use of $scCO_2$ can result in more intense colorizations. At supercritical conditions and constant pressure, increasing the pressure did not lead to higher dyeing intensities. At temperatures of 60 °C and 80 °C, dyestuff precipitated on the leather samples and changed the hue from red to shimmering orange.

On the basis of the results discovered in autoclave 1, to scale up in autoclave 2 we focused on a temperature of 40 °C. We considered the same pressure range of 80–250 bar and assessed the influence of the rotating basket.

Fig. 8 shows the values of the LCH color space depending on pressures between 80 bar and 250 bar at a temperature of 40 °C. In comparison with the trials in autoclave 1, the color space value trends concur. This means that by increasing the pressure, the C* values and H* values increase while the L* values decrease between 80 bar and 100 bar. A further increase in pressure does not lead to significant changes in the color values. We deduced that the properties of CO_2 at 100 bar and 40 °C in combination with the structure of the coating also lead to the maximum achievable dyeing intensities with the dye used in autoclave 2.

However, the LCH color value trends that resulted in autoclaves 1 and 2 are not identical. This is due to the higher volume of the autoclave and the rotation of the basket. At 80 bar and 40 °C, a higher C* value was achieved in autoclave 2 than in autoclave 1. This is due to temperature gradients at the beginning of the process. Despite pre-heating, the pressurization influences the temperatures inside the autoclaves. At the beginning of the process, the autoclaves are at ambient pressure and 40 °C, while the CO_2 is stored at 60 bar and 20 °C. In both machines, the CO_2 flow inside the autoclave causes a decrease in temperature due to the Joule-Thomson effect. The higher volume of autoclave 2 caused a decrease in temperature during pressurization that influenced the CO_2 properties, the dye concentration and thereby the colorization. In autoclave 1, the effect had less of an impact due to the lower volume.

We note that it was not possible to measure the temperature distribution within the autoclaves precisely. However, the SD of 11.4% for the C* value that resulted at 80 bar and 40 °C shows that the colorization is not uniform. This stresses the hypothesis of variating densities of CO_2 at the beginning of the process, which cause variating CO_2 properties and colorizations. Since the temperature gradient at the beginning of the process influences the dyeing result, the process time of 1 h is sufficient to equilibrate the system. The effect of the dyeing time will be considered in a further manuscript.

At 40 °C and pressures between 100 bar and 250 bar, the H* values that resulted from the experiments in autoclave 1 were about 22°. The higher H* values of 25–30° that resulted from the experiments in autoclave 2 were caused by unabsorbed dyestuff that was attached to the leather surface. H* values assigned to orange that resulted from

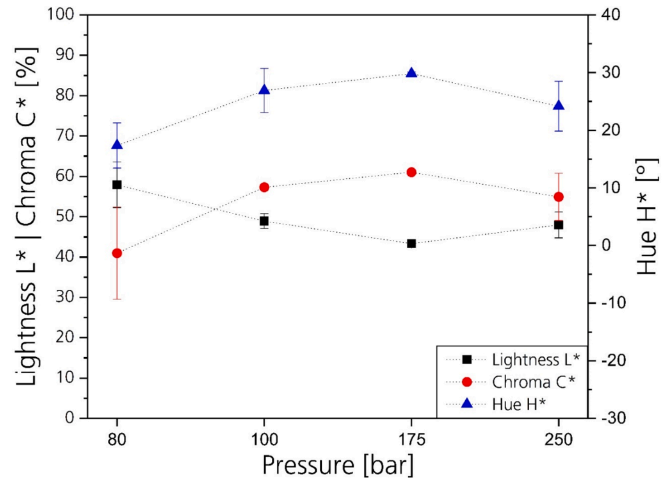

Fig. 8. Influence of the pressure on the LCH color values of the leather coating after $scCO_2$ dyeing at 40 °C.

autoclave 1 affiliate to dyestuff that was dissolved in CO_2 and precipitated on the leather surface. In the 5 L machine, we partly observed the same mechanism. The orange shimmering on the leather surface increases by increasing the pressure. This is due to the higher density and dye concentration. When the autoclave is depressurized, more dyestuff precipitates and is able to be adsorbed by the leather surface at 250 bar than for example at 100 bar. However, in the 5 L machine solid dyestuff in excess was mixed with the leather samples in the rotating basket during the dyeing procedure. This meant that solid, undissolved dyestuff could additionally contaminate the leather surface causing higher hue angles in comparison with autoclave 1. We found that it was practically impossible to remove the dyestuff from the surface by brushing, rubbing, vacuuming or other techniques. Since the dyestuff is insoluble in water and the polymer coating is not resistant to organic solvents, cleaning the surface with liquids was not possible either. An additional treatment with dense CO_2 led to discoloration of the samples. Therefore, the dyeing of leather at lower pressures with an optimized dyestuff amount seems to be beneficial to achieve optimal dyeing results.

Complementary to the color measurements, Fig. 9 shows micrographs of the leather surface after $scCO_2$ dyeing between 80 bar and 250 bar at 40 °C. The micrograph of the 80 bar trial shows a pale pink colorization. At pressures between 100 bar and 250 bar, similar red colorizations occurred. However, all samples partly show a shimmering effect. Areas of orange shimmering are visible in particular on the micrograph of the sample dyed at 250 bar.

In relation to the color measurements, the results of the micrographs shown in Fig. 9 were to be expected. The dyeing intensity of the samples dyed at 80 bar is much lower compared with the samples dyed at higher pressures. Contrary to this, the dyeing intensity between the samples dyed at pressures between 100 bar and 250 bar concur with the corresponding LCH color values.

In summary, the optimal outcome regarding color uniformity and intensity at the lowest possible pressure and temperature results in a pressure of 100 bar and a temperature of 40 °C.

3.2.2. Influence of the loading and dyestuff amount

The aim of the test series was to identify usage of dyestuff that allows intense colorizations (C* values of 50% and H* values of 20°) by avoiding excess dyestuff. In addition to the previous test series, we assessed the rub fastness of the dyed leather samples as described in Section 2.2.4.

Independent of the loading, all samples from each trial were uniformly dyed. However, the usage of dyestuff to achieve a chroma of 50% differed considerably depending on the loading. Figs. 10–12 show the LCH values of the dyed leather samples depending on the loading and

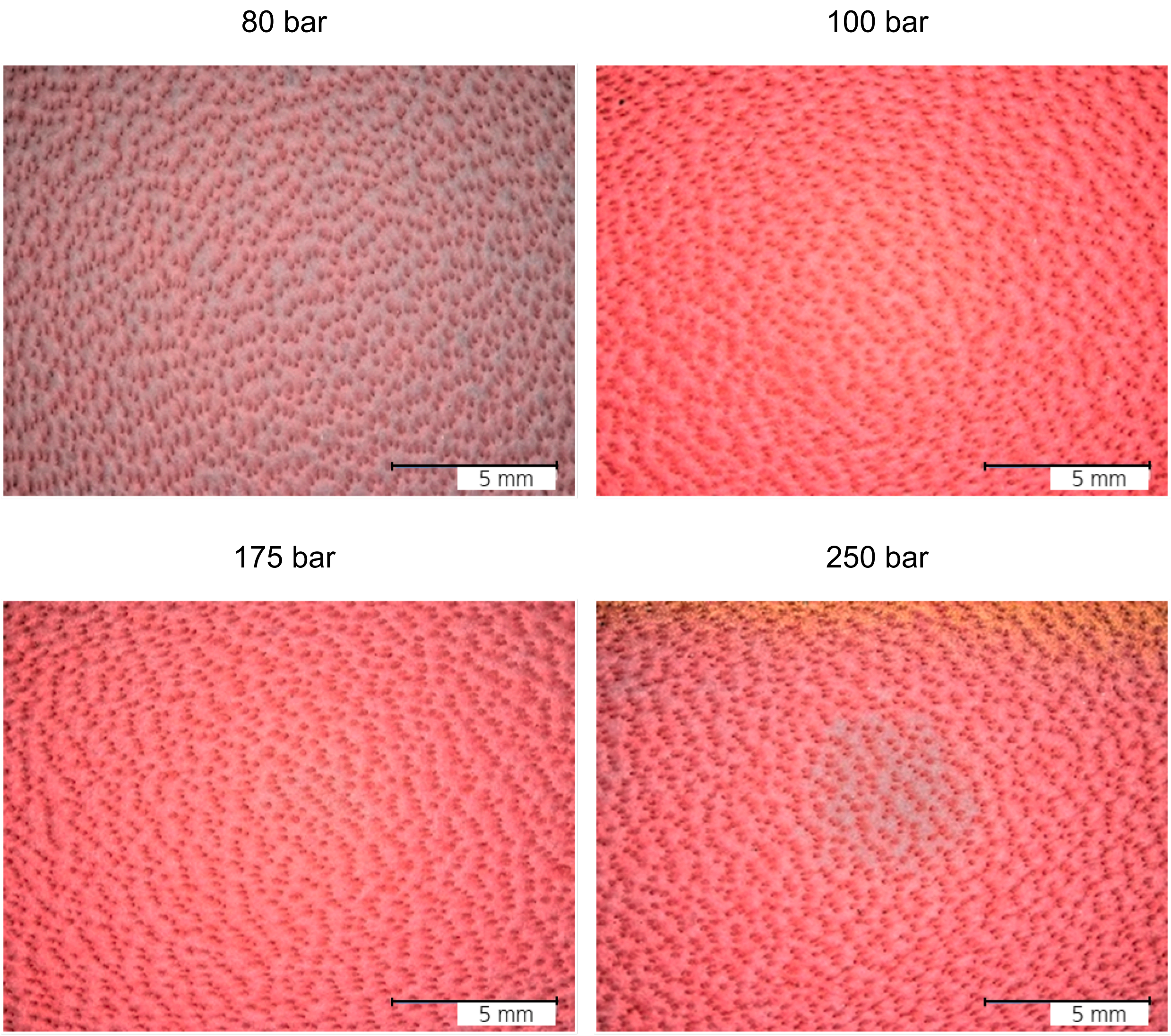

Fig. 9. Micrographs of leather surfaces dyed using $scCO_2$ at 40 °C and pressures between 80 bar and 250 bar.

usage of dyestuff per ft^2 leather. The trends in the diagrams show that the dyeing intensities increased by increasing the usage of dyestuff. However, we observed that by increasing the load, the quantity of dyestuff needed in relation to the leather area to achieve similar colorizations decreased. At a load of 3 samples, 0.36 g/ft^2, at a load of 15 samples, 0.14 g/ft^2 and at a load of 30 samples, 0.1 g/ft^2 of dyestuff are needed to achieve C* values of about 50% and H* values of 20°.

Table 5 shows the results of the rub fastness tests depending on the load and usage of dyestuff. Independent of the load and the usage of dyestuff, all samples show the maximum achievable rub fastness grades of 5 regardless of dry or wet condition. Considering the results of the perspiration felts, rub fastness grades of 5 are also reached, though not at the highest dyeing intensities. At the minimum dyestuff usages that result in C* values of 50%, the grades decrease to 4 or 4.5. By increasing the usage of dyestuff, the rub fastness grades decrease to 3.5.

Fig. 13 shows micrographs of dyed leather samples with a dyestuff usage of 0.1 g/ft^2 depending on a load of 3, 15 and 30 samples to visualize the influence of the loading. The correlation between the load and the dyeing intensities at the same dyestuff usage per ft^2 is mainly due to the amount of dyestuff that can be dissolved in CO_2 and the partitioning of the dyestuff between CO_2 and the polymeric coating. At 100 bar and 40 °C, about 70 mol of CO_2 are available in autoclave 2. Under these conditions, about 0.02 g of dyestuff are needed to saturate CO_2 with dyestuff [37]. At a load of 30 samples that correspond to about

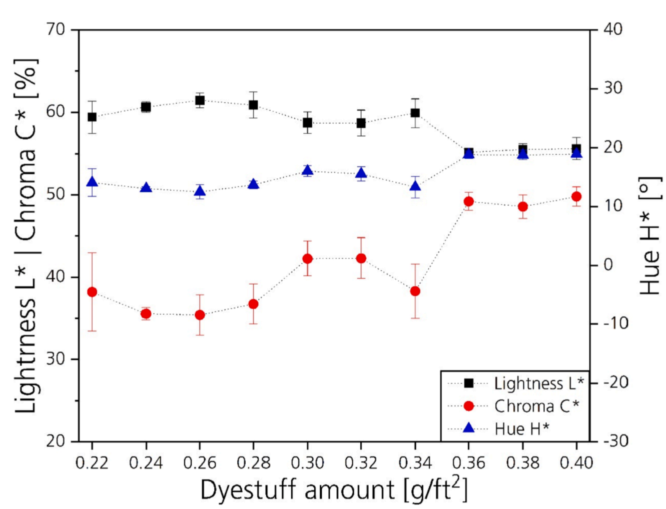

Fig. 10. Influence of the dyestuff amount on the LCH color space after $scCO_2$ dyeing at 100 bar and 40 °C at a load of 3 leather samples.

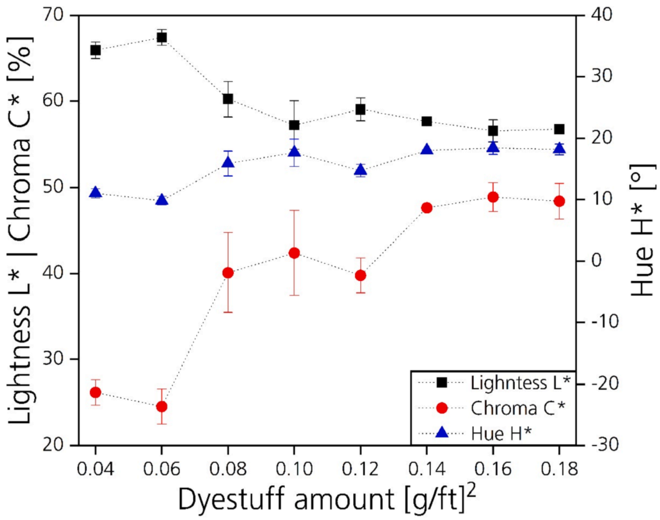

Fig. 11. Influence of the dyestuff amount on the LCH color space after $scCO_2$ dyeing at 100 bar and 40 °C at a load of 15 leather samples.

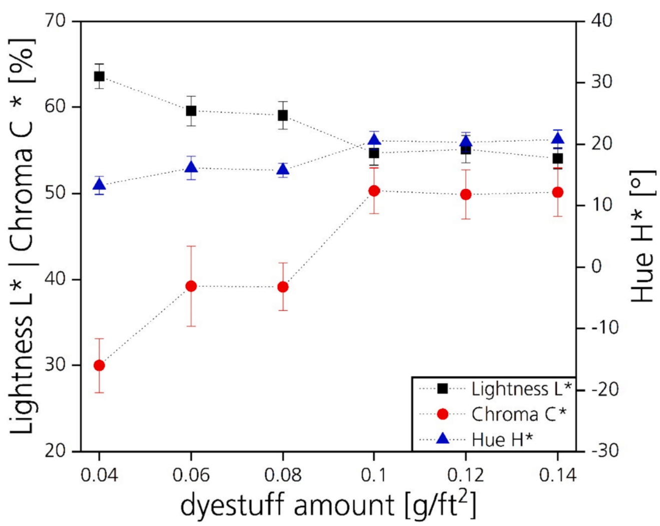

Fig. 12. Influence of the dyestuff amount on the LCH color space after $scCO_2$ dyeing at 100 bar and 40 °C at a load of 30 leather samples.

2 ft^2, 0.2 g of dyestuff were added to achieve the highest dyeing intensity. This equates to 0.1 g of dye per ft^2 of leather. Since a maximum of 0.02 g can be dissolved in CO_2, a maximum of 10% of the 0.2 g of dyestuff remains dissolved in CO_2. The rub fastness tests showed that there are poorly precipitated amounts of excess dyestuff on the leather surface. This indicates that most of the residual 90% of the dye used was absorbed by the leather coating. By decreasing the load, the portion of dyestuff used that can remain in CO_2 increases, leading to lower dyeing intensities at the same dyestuff usage per ft^2.

Table 5
Rub fastness test results of $scCO_2$-dyed leather samples at 100 bar and 40 °C depending on the load and quantity of dyestuff.

Load [pcs]	Dye quantity [g/ft^2]	Condition of felt specimen		
		Dry	Wet	Perspiration
3	0.34	5	5	5
	0.36	5	5	4
	0.38	5	5	4
	0.40	5	5	3.5
15	0.12	5	5	5
	0.14	5	5	4.5
	0.16	5	5	4
	0.18	5	5	3.5
30	0.08	5	5	5
	0.10	5	5	4.5
	0.12	5	5	4
	0.14	5	5	3.5

3.2.3. *Study of scalability*

The previous test series showed that intense colorization in combination with the highest rub fastness grades are possible by minimizing excess dyestuff. However, it was found that the amount of dye dissolvable in CO_2 can influence the dyeing intensity when the dye usage is calculated in relation to the leather area. By decreasing the load, the dyeing intensities decrease at the same usage of dye per ft^2 of leather. However, the volume of the autoclave was not exhausted.

We performed a trial with 6 ft^2 of leather to assess the scalability and flexibility of the process. At 6 ft^2, about 50% of the volume of the basket was filled with leather samples. This loading ensured that the samples were able to mix while rotating during the process. On the basis of previous results, the dyeing process was performed at 100 bar and 40 °C for 1 h with 0.6 g of dyestuff (0.1 g dye per ft^2 of leather).

Fig. 14 shows a photograph of the resulting 90 uniformly dyed leather samples at an area of 6 ft^2, which were processed in a single procedure in the 5 L autoclave. The average LCH color space values ($C^* = 53.4\%$, $L^* = 55.2\%$, $H^* = 20.3°$) were comparable with the LCH values of the 2 ft^2 trial in the previous test series. Rub fastness grades of 5 resulted for dry and wet pads. The perspiration tests showed slight discoloration of the pads resulting in grades of 4.5.

Since the deviations in the LCH color space values of the dyeing experiment using 30 samples in the previous test series and of the 90-sample experiment are negligible, these different loadings have no influence on the colorization. This shows that the influence of the loading is negligible when the ratio of the dyestuff that can be dissolved in CO_2 to the dyestuff that is absorbed by the coating is in favor of the absorbable portion.

At 2 ft^2 of leather, a maximum of 10% of the dye used remained dissolved in CO_2, while at 6 ft^2 and a dye usage of 0.6 g, a maximum of 3% remained dissolved in CO_2. The rest of the dye diffuses in the leather coating or remained in the autoclave as excess dyestuff. Considering the high rub fastness grades, we assume that the amount of excess dyestuff is negligible and the dye exhaustion is about 97% at a load of 50%. It can be assumed that, in dyeing machines designed for the colorization of leather, much higher loadings than 50% are possible and therefore dye exhaustions of nearly 100% can be realized.

4. Conclusion

We performed CO_2 dyeing of leather by impregnating the leather coating with a red dyestuff in two autoclaves with volumes of 63 mL and 5 L.

The trials in autoclave 1 (63 mL) showed that colorization of the leather samples in subcritical and supercritical conditions is possible. However, $scCO_2$ dyeing results in more intense colorization. This is mainly due to better diffusivity of the supercritical phase in comparison with the subcritical phase. The optimal outcome regarding dyeing intensity at the lowest possible pressure and temperature results in a pressure of 100 bar and a temperature of 40 °C. Higher pressures of up to 250 bar and higher temperatures of up to 80 °C do not improve the colorization. Compared with $scCO_2$ polyester dyeing, which operates at pressures of above 200 bar and 100 °C, the energy efficiency of the $scCO_2$ leather dyeing process is improved. The reason for this is the structure of the leather coating, which allows a high dye uptake under comparable moderate dyeing conditions.

In autoclave 2 (5 L), the process conditions of 100 bar and 40 °C resulted in LCH values of 55% (lightness), 50% (chroma) and 20° (hue angle). These values show uniformly intense colorization and are in the

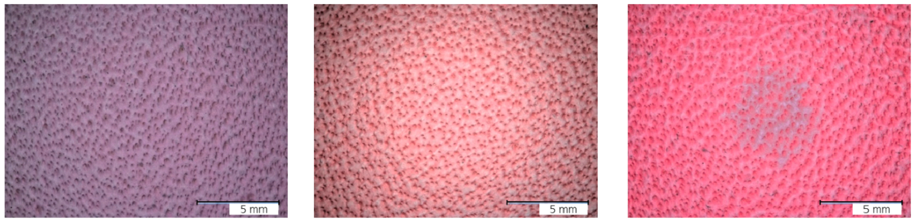

Fig. 13. Micrographs of coated leather samples after $scCO_2$ dyeing at 100 bar and 40 °C with 0.1 g of dye per ft^2 at loadings of 3 (left), 15 (middle) and 30 samples (right).

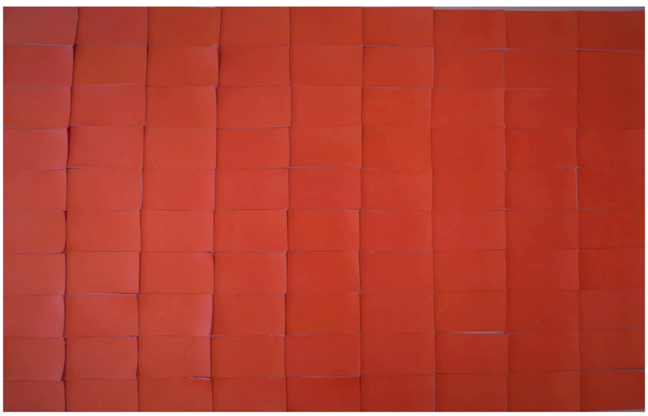

Fig. 14. Photograph of 90 leather samples with a total area of 6 ft^2 after $scCO_2$ dyeing in a 5 L autoclave at 100 bar and 40 °C for 1 h.

given range of the dye supplier for several polymers.

The amount of dyestuff needed to achieve the most intense colorization without adsorbed excess dyestuff on the leather surface depends on the loading. At loadings below 20%, additional dyestuff has to be added to saturate the CO_2 during the dyeing procedure. By comparing loadings of 20% and 50%, the quantity of excess dye is negligible. It is assumed that this observation can be transferred to higher loadings.

By dyeing 6 ft^2 of leather, we used 0.6 g (0.1 g/ft^2) to achieve an optimal dyeing result. In conventional spray coating, about 1 g of dye or pigment is used per ft^2. This means that about 90% of the dyestuff can be saved. Furthermore, chemical agents as surfactants and formic acid are not needed. In conventional spray coating, up to 40% of the dyes and pigments are lost with the wastewater due to over-spraying. Since the production of dyes has an enormous negative effect on the sustainability of tanneries, the saving potential of the $scCO_2$ dyeing process is a useful approach to reduce the environmental burden and save on costs for dyes, pigments and other chemical agents.

The assessment of the rub fastness results is a special feature of the process. After conventional water-based leather dyeing, rub fastness grades of 3 or less are not unusual. For automotive or furniture leather, rub off is avoided by coating the colorized leather with a transparent finish. The $scCO_2$ dyeing process has the potential to dye leather without the need for a transparent finish. This feature may have two benefits: The transparent finish makes the coating thicker, leading to less natural haptics. Without the finish, the tanner would be able to produce colored leather with new haptics. The second benefit is the possibility that full leathers could be produced and individually $scCO_2$-dyed at the end of the process. This would mean that the $scCO_2$ leather dyeing process could shorten the time between a customer request for a specific color and production of the corresponding leather by the tannery.

In conclusion, the $scCO_2$ leather dyeing process offers significant potential to avoid contaminated wastewater and reduce costs for dyestuff, chemical agents and wastewater treatment. This manuscript presents the first promising results in the field of $scCO_2$ leather dyeing. Several questions regarding the flexibility of colors, ability to dye other coatings and, in particular, dyeing of the full cross-section have to be answered in future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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